UNIT 2 · ARS/NET PATTERN

# MCQ Bank — 215 Questions with Answers

## Machinery elements — Topics 1–2 (Q1–Q40)

**Q1. Power transmitted by a shaft is given by:**

(a) 2πNT/60 W (b) πNT/60 W
(c) NT/60 W (d) 2NT/π W

**Q2. In a belt drive, the ratio $T_1/T_2$ equals:**

(a) e^kθ (b) kθ
(c) e^θ/k (d) log(kθ)

**Q3. The tension ratio commonly assumed in designing V-belt drives is:**

(a) 2 (b) 3
(c) 5 (d) 8

**Q4. For rubber flat belts on steel pulleys, the tension ratio at 180° arc of contact is about:**

(a) 1.2–1.5 (b) 2.2–2.6
(c) 4–5 (d) 6–8

**Q5. The included angle of a standard V-belt cross-section is:**

(a) 30° (b) 34°
(c) 38° (d) 40°

**Q6. Sheave groove angles are kept at 34–38°, i.e., less than the belt angle, to:**

(a) Reduce belt speed (b) Ensure wedging action
(c) Increase slip (d) Ease belt removal

**Q7. A V-belt sheave and a flat pulley are balanced in power capacity when arcs of contact are about:**

(a) 180° and 180° (b) 130° and 230°
(c) 230° and 130° (d) 90° and 270°

**Q8. Belt power is maximum when centrifugal tension equals:**

(a) $T_1$ (b) $T_1/2$
(c) $T_1/3$ (d) $3T_1$

**Q9. Creep in belts is caused by:**

(a) Overload slip (b) Alternate elastic elongation/contraction of the belt over pulleys
(c) Misalignment (d) Wear

**Q10. Agricultural V-belt cross-sections per ASAE standards are designated:**

(a) A, B, C (b) HA, HB, HC, HD, HE
(c) SPA, SPB (d) 3V, 5V, 8V

**Q11. Double (hexagonal) V-belts are used for:**

(a) Higher speeds (b) Serpentine drives with reverse bends
(c) Vertical shafts (d) Wet conditions

**Q12. Flat-belt pulleys are crowned to:**

(a) Increase friction
(b) Keep the belt tracking at pulley centre
(c) Reduce weight
(d) Balance the pulley

**Q13. An idler pulley placed on the slack side near the small pulley:**

(a) Reduces arc of contact
(b) Increases arc of contact and capacity
(c) Increases slip
(d) Reduces belt life only

**Q14. Variable-speed V-belt drives change speed by:**

(a) Changing motor speed
(b) Adjustable-pitch sheaves altering effective pitch diameter
(c) Gearbox
(d) Slipping clutch

**Q15. Chordal (polygonal) action in chain drives causes:**

(a) Chain elongation
(b) Cyclic variation in chain velocity
(c) Wear of pins
(d) Sprocket eccentricity

**Q16. To minimize chordal action, the smaller sprocket should generally have at least:**

(a) 6 teeth
(b) 10 teeth
(c) 17 teeth
(d) 30 teeth

**Q17. The chain best suited to gritty, slow-speed farm conveyor service and field repair is:**

(a) Silent chain
(b) Precision roller chain
(c) Detachable-link (steel/malleable) chain
(d) Leaf chain

**Q18. Chain speed (m/s) with pitch p (m), sprocket teeth T, speed N (rpm) is:**

(a) pTN/60
(b) pT/60N
(c) 60pTN
(d) pN/60T

**Q19. Unlike belts, chain drives are operated with:**

(a) High initial tension
(b) A small amount of slack
(c) Zero backlash
(d) Crowned sprockets

**Q20. Attachment links (A- and K-type) on roller chains are used in agriculture to:**

(a) Join chain ends
(b) Mount slats/paddles on conveyor/elevator chains
(c) Reduce noise
(d) Increase strength

**Q21. The standard pressure angle for involute gears today is:**

(a) 14.5°
(b) 20°
(c) 25°
(d) 30°

**Q22. Minimum number of teeth on a 20° full-depth pinion to avoid interference is about:**

(a) 12
(b) 17
(c) 25
(d) 32

**Q23. Module of a gear equals:**

(a) T/D
(b) D/T
(c) πD/T
(d) T/πD

**Q24. The gear pair connecting shafts whose axes are offset (non-intersecting, non-parallel), used in automotive/tractor drive lines, is:**

(a) Spiral bevel
(b) Hypoid
(c) Herringbone
(d) Rack and pinion

**Q25. A worm-and-wheel drive is preferred where:**

(a) High efficiency is needed
(b) Large speed reduction in one stage (and possible self-locking) is needed
(c) Shafts are parallel
(d) Axial thrust must be zero

**Q26. In the Lewis equation $F_t = \sigma \cdot b \cdot Y \cdot m$, Y is the:**

(a) Wear factor
(b) Lewis form factor (tooth shape)
(c) Service factor
(d) Velocity factor

**Q27. An idler gear in a simple gear train:**

(a) Changes speed ratio
(b) Changes only the direction of rotation
(c) Increases torque
(d) Reduces noise

**Q28. $L_{10}$ ($B_{10}$) bearing life is the life that:**

(a) 10% of bearings complete
(b) 90% of a group of bearings will complete or exceed
(c) Average bearing achieves
(d) The best bearing achieves

**Q29. Bearing life varies with load as:**

(a) $L \propto C/P$
(b) $L \propto (C/P)^3$ for ball bearings
(c) $L \propto (C/P)^2$
(d) L ∝ e^(C/P)

**Q30. For combined heavy radial and axial loads in tractor wheel hubs, the bearing used is:**

(a) Deep-groove ball
(b) Needle roller
(c) Tapered roller
(d) Self-aligning ball

**Q31. Universal-joint cross trunnions usually run in:**

(a) Bush bearings
(b) Needle roller bearings
(c) Ball bearings
(d) Tapered rollers

**Q32. Speed fluctuation of a single Hooke's joint is eliminated in a PTO shaft by:**

(a) Higher speed
(b) Two joints with in-phase yokes and equal joint angles
(c) Heavier shaft
(d) Lubrication

**Q33. Oldham coupling accommodates:**

(a) Angular misalignment
(b) Parallel (lateral) shaft offset
(c) Axial float only
(d) Torsional vibration

**Q34. The overload device that gives an audible warning and re-engages automatically is the:**

(a) Shear pin
(b) Jump (ratchet) clutch
(c) Rigid coupling
(d) Key

**Q35. An overrunning (freewheel) clutch in a rotavator/PTO driveline:**

(a) Limits torque
(b) Allows the high-inertia rotor to overrun the PTO while slowing
(c) Reverses drive
(d) Increases speed

**Q36. Recommended spring index (C = D/d) for helical springs is:**

(a) 1–3
(b) 6–12
(c) 15–25
(d) 30–50

**Q37. Wahl factor corrects the stress in helical springs for:**

(a) Temperature
(b) Curvature and direct shear effects
(c) Buckling
(d) Fatigue only

**Q38. A bolt of property class 8.8 has minimum ultimate tensile strength and yield strength of:**

(a) 800 and 640 MPa
(b) 880 and 800 MPa
(c) 800 and 880 MPa
(d) 640 and 800 MPa

**Q39. The 540-rpm tractor PTO shaft has:**

(a) 21 splines
(b) 6 splines
(c) 8 splines
(d) 10 splines

**Q40. The self-aligning key seated in a semicircular shaft recess is the:**

(a) Feather key
(b) Kennedy key
(c) Woodruff key
(d) Gib-head key

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**Answers**

Q1 — (a) 2πNT/60 W

Q2 — (a) e^kθ

Q3 — (c) 5 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°; V-belt design assumes R = 5 (k ≈ 0.51 including wedging).*

Q4 — (b) 2.2–2.6 — *Kepner §4.3 — flat rubber belt k = 0.25–0.30 gives R = 2.2–2.6 at 180°.*

Q5 — (d) 40°

Q6 — (b) Ensure wedging action

Q7 — (b) 130° and 230° — *Equal kθ values → 130° sheave ≈ 230° flat pulley (Kepner §4.9); basis of V-flat drives on threshers.*

Q8 — (c) $T_1/3$ — *dP/dv = 0 with P = ($T_1$ -T_c-$T_2$')v … yields T_c = $T_1$/3.*

Q9 — (b) Alternate elastic elongation/contraction of the belt over pulleys

Q10 — (b) HA, HB, HC, HD, HE

Q11 — (b) Serpentine drives with reverse bends

Q12 — (b) Keep the belt tracking at pulley centre

Q13 — (b) Increases arc of contact and capacity

Q14 — (b) Adjustable-pitch sheaves altering effective pitch diameter

Q15 — (b) Cyclic variation in chain velocity

Q16 — (c) 17 teeth — *Chordal velocity variation ∝ (1 - cos 180°/T); below 17 teeth the fluctuation and noise rise steeply.*

Q17 — (c) Detachable-link (steel/malleable) chain

Q18 — (a) pTN/60

Q19 — (b) A small amount of slack

Q20 — (b) Mount slats/paddles on conveyor/elevator chains

Q21 — (b) 20°

Q22 — (b) 17 — *Minimum teeth = 2/sin²20° ≈ 17.1 → 17–18.*

Q23 — (b) D/T

Q24 — (b) Hypoid

Q25 — (b) Large speed reduction in one stage (and possible self-locking) is needed

Q26 — (b) Lewis form factor (tooth shape)

Q27 — (b) Changes only the direction of rotation

Q28 — (b) 90% of a group of bearings will complete or exceed — *$L_{10}$ = (C/P)^p million rev; 90% reliability definition — repeatedly asked.*

Q29 — (b) L ∝ $(C/P)^3$ for ball bearings — *$L_{10}$ = (C/P)^p million rev, with p = 3 for ball bearings.*

Q30 — (c) Tapered roller

Q31 — (b) Needle roller bearings

Q32 — (b) Two joints with in-phase yokes and equal joint angles — *PTO drivelines are double-Cardan arrangements; equal angles + in-phase yokes cancel the cos α fluctuation.*

Q33 — (b) Parallel (lateral) shaft offset
Q34 — (b) Jump (ratchet) clutch
Q35 — (b) Allows the high-inertia rotor to overrun the PTO while slowing
Q36 — (b) 6–12
Q37 — (b) Curvature and direct shear effects
Q38 — (a) 800 and 640 MPa — *Class X.Y → UTS = 100X MPa; yield = 0.Y × UTS.*
Q39 — (b) 6 splines
Q40 — (c) Woodruff key

## System characteristics, balancing and stability — Topics 3–4 (Q41–Q75)

**Q41. The capacity of a multi-component machine system (e.g., combine) is determined by:**

(a) The largest component
(b) The limiting (weakest) functional component
(c) The engine alone
(d) Average of all components

**Q42. If a machine has 4 components in series, each with reliability 0.95, system reliability is about:**

(a) 0.95
(b) 0.99
(c) 0.81
(d) 0.86

**Q43. Draft of a mouldboard plough varies with forward speed as:**

(a) $D = a + bV$
(b) $D = a + bV^2$
(c) $D = a/V$
(d) D = constant

**Q44. Quality of work of a seed drill is judged by:**

(a) Fuel use
(b) Seed placement uniformity/depth accuracy
(c) Weight
(d) Draft only

**Q45. RNAM test codes relate to:**

(a) Tractor engines
(b) Standard test procedures for agricultural machines in Asia-Pacific region
(c) Chemical residues
(d) Irrigation design

**Q46. A rotor at rest in any angular position without turning is:**

(a) Dynamically balanced
(b) Statically balanced
(c) Unbalanced
(d) Over-balanced

**Q47. Static balance requires:**

(a) ΣForces = 0
(b) ΣCouples = 0
(c) Both zero
(d) High speed

**Q48. Dynamic balance additionally requires:**

(a) Zero resultant couple
(b) Higher mass
(c) One correction plane
(d) Rigid shaft

**Q49. Masses balanced in one plane but distributed along the shaft may still produce:**

(a) No effect
(b) A rocking (unbalanced) couple
(c) Static unbalance
(d) Torque rise

**Q50. Centrifugal force due to unbalance varies as:**

(a) $\omega$
(b) $\omega^2$
(c) $\omega^3$
(d) $\sqrt{\omega}$

**Q51. Doubling the speed of an unbalanced thresher cylinder increases the unbalance force:**

(a) 2×
(b) 4×
(c) 8×
(d) No change

**Q52. Complete dynamic balancing of rotating masses in several planes needs:**

(a) One counterweight
(b) Correction in two planes
(c) Flywheel
(d) Damper

**Q53. A reciprocating inertia force CANNOT be fully balanced by a rotating counterweight because:**

(a) It is too large
(b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle
(c) Speeds differ
(d) Friction interferes

**Q54. In mower balancing practice, the rotating counterweight is sized to balance the rotating parts plus about:**

(a) One-fourth
(b) One-half of maximum reciprocating force
(c) Full reciprocating force
(d) Twice the reciprocating force

**Q55. In the analysis of mower pitman forces, the pitman mass is idealized as:**

(a) Uniform rod
(b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)
(c) Point at centre
(d) Neglected

**Q56. The primary reciprocating inertia force has a frequency equal to:**

(a) Half crank speed
(b) Crank speed
(c) Twice crank speed
(d) Four times

**Q57. Two opposed reciprocating counter-masses fully cancel forces only if they are:**

(a) Unequal
(b) Collinear (otherwise a vibratory couple results)
(c) At right angles
(d) On separate shafts

**Q58. The double-knife (opposed knives) cutter bar permits:**

(a) Lower cutting speed
(b) Half stroke per knife and higher crank rpm → higher forward speed
(c) Heavier knives
(d) No guards in all versions

**Q59. In a dynamically balanced mower (Kepner Fig. 14.6), the balancing element is:**

(a) Heavier flywheel
(b) Reciprocating counterweight on the inner shoe
(c) Rubber mounts
(d) Longer pitman

**Q60. The energy-smoothing member for the baler plunger's cyclic load is the:**

(a) Knotter
(b) Flywheel
(c) Pickup
(d) Bale tension rail

**Q61. Coefficient of fluctuation of speed is:**

(a) $(\omega_{max}-\omega_{min})/\omega_{mean}$
(b) $\omega_{mean}/(\omega_{max}-\omega_{min})$
(c) $\omega_{max}/\omega_{min}$
(d) $\Delta\omega^2$

**Q62. Required flywheel moment of inertia for energy fluctuation ΔE is:**

(a) $\Delta E \cdot C_s \cdot \omega^2$
(b) $\Delta E/(C_s \cdot \omega^2_{mean})$
(c) $C_s/\Delta E$
(d) $\frac{1}{2}I\omega^2$

**Q63. Resonance occurs when:**

(a) Damping is high
(b) Excitation frequency equals natural frequency
(c) Mass is large
(d) Speed is low

**Q64. Natural frequency of a spring-mass system is:**

(a) √(m/k) (b) √(k/m)

(c) k/m (d) m/k

**Q65. Vibration isolation (transmissibility <1) requires frequency ratio greater than:**

(a) 1 (b) √2

(c) 2 (d) π

**Q66. A machine on a side slope overturns when the vertical through the CG:**

(a) Passes inside wheelbase (b) Falls outside the wheel-contact polygon

(c) Is at axle height (d) Coincides with draft line

**Q67. Lateral overturn limit angle with track width B and CG height h is given by:**

(a) tan θ = 2h/B (b) tan θ = B/2h

(c) sin θ = B/h (d) θ = B·h

**Q68. Lateral stability is improved by:**

(a) Higher CG, narrow track (b) Lower CG, wider track

(c) More speed (d) Higher hitch

**Q69. With a heavy rear-mounted implement raised, safe steering requires the front axle to retain at least about:**

(a) 5% (b) 20%

(c) 40% (d) 60% of total weight

**Q70. Front-end ballast requirement for a mounted implement is computed by:**

(a) Energy balance (b) Moment balance about the rear axle

(c) Draft equation (d) Traction equation

**Q71. Positive tongue (hitch) load recommended on trailed trailers is about:**

(a) 0% (b) 10–15% of gross weight

(c) 30–40% (d) 50%

**Q72. Snaking/weaving of a trailed implement at transport speed is reduced by:**

(a) Negative tongue load (b) Adequate tongue load and proper hitch length

(c) Higher speed (d) Wider tyres

**Q73. An unbalanced high-speed cylinder in a thresher primarily causes:**

(a) Better threshing (b) Vibration, bearing failure and frame cracks

(c) Higher capacity (d) Less noise

**Q74. Timeliness of operation in machinery system evaluation refers to:**

(a) Machine speed (b) Completing the operation within the optimum period to avoid yield/quality loss

(c) Idle time (d) Shift hours

**Q75. MTBF stands for:**

(a) Maximum time before failure (b) Mean time between failures

(c) Minimum test bench force (d) Mean torque at base frequency

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**Answers**

Q41 — (b) The limiting (weakest) functional component

Q42 — (c) 0.81 — *$0.95^4 = 0.8145$ — series reliability multiplies; why simple, robust designs matter in farm machines.*

Q43 — (b) $D = a + bV^2$

Q44 — (b) Seed placement uniformity/depth accuracy

Q45 — (b) Standard test procedures for agricultural machines in Asia-Pacific region
Q46 — (b) Statically balanced
Q47 — (a) ΣForces = 0
Q48 — (a) Zero resultant couple
Q49 — (b) A rocking (unbalanced) couple
Q50 — (b) ω²
Q51 — (b) 4× — *F = mrω² → 2² = 4×.*
Q52 — (b) Correction in two planes
Q53 — (b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle — *Kepner §14.11 — the rotating counterweight force is constant at all crank angles.*
Q54 — (b) One-half of maximum reciprocating force — *Kepner §14.11 — compromise counterweight = F_c + ½F_r,max ("half balancing"), residual unbalance shifted 90°.*
Q55 — (b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)
Q56 — (b) Crank speed
Q57 — (b) Collinear (otherwise a vibratory couple results)
Q58 — (b) Half stroke per knife and higher crank rpm → higher forward speed — *Kepner §14.12 — each knife stroke halves; crank speed (and forward speed) can nearly double for the same inertia stress.*
Q59 — (b) Reciprocating counterweight on the inner shoe
Q60 — (b) Flywheel
Q61 — (a) (ω_max-ω_min)/ω_mean
Q62 — (b) ΔE/(C_s·ω²_mean)
Q63 — (b) Excitation frequency equals natural frequency
Q64 — (b) √(k/m)
Q65 — (b) √2 — *Transmissibility < 1 only when ω/ω_n > √2 — basis of soft engine mounts.*
Q66 — (b) Falls outside the wheel-contact polygon
Q67 — (b) tan θ = B/2h — *At tipping, the CG vertical passes through the downhill wheel contact: tan θ = (B/2)/h.*
Q68 — (b) Lower CG, wider track
Q69 — (b) 20%
Q70 — (b) Moment balance about the rear axle
Q71 — (b) 10–15% of gross weight
Q72 — (b) Adequate tongue load and proper hitch length
Q73 — (b) Vibration, bearing failure and frame cracks
Q74 — (b) Completing the operation within the optimum period to avoid yield/quality loss
Q75 — (b) Mean time between failures

## Force analysis, pull–draft–power and furrow openers — Topics 5–6 (Q76–Q110)

**Q76. Draft of an implement is:**

(a) Total force along the hitch line
(b) Horizontal component of pull in the direction of travel
(c) Vertical component of pull
(d) Side component of pull

**Q77. A pull of 3000 N acts at 20° above horizontal. The draft is about:**

(a) 1026 N
(b) 2819 N
(c) 3000 N
(d) 3192 N

**Q78. Unit draft is defined as draft per unit:**

(a) Width of implement
(b) Weight of implement
(c) Cross-sectional area of furrow
(d) Travel speed

**Q79. The unit of unit draft is:**

(a) N
(b) N/cm
(c) N/cm²
(d) N·m

**Q80. 1 kgf/cm² is approximately:**

(a) 9.81 kPa
(b) 98.1 kPa
(c) 981 kPa
(d) 9810 kPa

**Q81. A 2-bottom MB plough (30 cm/bottom) works 15 cm deep with unit draft 0.6 kgf/cm². Total draft is:**

(a) 270 kgf
(b) 405 kgf
(c) 540 kgf
(d) 810 kgf

**Q82. The plough in Q6 moves at 1.5 m/s. Drawbar power is about:**

(a) 5.4 hp
(b) 7.2 hp
(c) 10.8 hp
(d) 14.4 hp

**Q83. With D in kgf and v in km/h, drawbar power in kW equals:**

(a) D·v/75
(b) D·v/270
(c) D·v/367
(d) D·v/1000

**Q84. Draft 9 kN at 1.5 m/s gives drawbar power:**

(a) 6 kW
(b) 9 kW
(c) 13.5 kW
(d) 60 kW

**Q85. In Goryachkin's equation P = fG + kab + εabv², the speed-dependent term is:**

(a) fG
(b) kab
(c) εabv²
(d) All three

**Q86. The term fG in Goryachkin's equation represents:**

(a) Soil cutting resistance
(b) Friction/rolling resistance due to implement weight
(c) Soil throwing
(d) Suction force

**Q87. If ploughing speed is doubled, the dynamic (soil-throwing) component of draft becomes:**

(a) 2×
(b) 4×
(c) 8×
(d) Unchanged

**Q88. In ASAE D497 the soil-texture factor Fi for a mouldboard plough in fine (clay) soil is:**

(a) 0.45
(b) 0.70
(c) 1.0
(d) 1.5

**Q89. The centre of resistance of an MB plough bottom is commonly assumed at:**

(a) ¼ width from landside, ½ depth
(b) ½ width, full depth
(c) At the share point
(d) At the heel of landside

**Q90. For steady running with minimum draft, the line of pull must pass through:**

(a) The rear axle
(b) Centre of resistance and hitch point
(c) The plough heel
(d) The mast top

**Q91. Side draft occurs when:**

(a) Speed is high
(b) The line of pull is laterally offset from the centre of resistance
(c) Depth is uniform
(d) Soil is sandy

**Q92. Hitching a trailed plough too HIGH on the tractor drawbar generally causes:**

(a) Deeper penetration
(b) Reduced penetration (plough runs shallow)
(c) No change
(d) More side draft only

**Q93. A disc plough penetrates mainly because of:**

(a) Suction of the bottom
(b) Added weight/ballast
(c) Sharp landside
(d) High speed

**Q94. Which of these is a PARASITIC force on a plough?**

(a) Soil cutting resistance
(b) Furrow slice acceleration
(c) Landside friction
(d) Soil inversion resistance

**Q95. In L–V–S force analysis of a mounted implement, V denotes:**

(a) Velocity
(b) Vertical force component
(c) Volume of soil
(d) Side force

**Q96. Draft of a TRAILED implement is best measured with:**

(a) Torque meter on PTO
(b) Dynamometer between drawbar and implement
(c) Fuel meter
(d) Tachometer

**Q97. Draft of a MOUNTED implement is measured with:**

(a) Spring balance on top link only
(b) Three-point hitch (strain-gauge) dynamometer
(c) Rope and pulley
(d) Prony brake

**Q98. Compared with sandy soil, the unit draft of ploughing heavy clay is:**

(a) Lower
(b) Same
(c) Higher
(d) Zero

**Q99. Draft of a tillage implement depends on:**

(a) Soil type and moisture
(b) Depth and width of cut
(c) Speed and tool geometry
(d) All of these

**Q100. Draft of seed drills and harrows is usually expressed per unit:**

(a) Furrow area
(b) Working width
(c) Weight
(d) Wheel diameter

**Q101. Drawbar power equals engine power multiplied by:**

(a) Transmission efficiency only
(b) Tractive efficiency only
(c) Transmission × tractive efficiency
(d) Field efficiency

**Q102. Engine power 60 kW, transmission efficiency 0.90, tractive efficiency 0.67. Drawbar power ≈**

(a) 24 kW
(b) 30 kW
(c) 36 kW
(d) 54 kW

**Q103. The furrow opener best suited for zero-till sowing into anchored rice residue is:**

(a) Runner
(b) Inverted-T
(c) Shovel
(d) Full-width sweep

**Q104. Double-disc furrow openers are preferred on planters because they give:**

(a) Maximum soil inversion
(b) Precise, uniform seed placement and cut trash
(c) Deepest furrows
(d) Lowest cost

**Q105. Hoe-type furrow openers are best suited to:**

(a) Wet puddled soil
(b) Hard and dry soils
(c) Loose sand only
(d) Standing water

**Q106. Runner-type openers require:**

(a) Trashy uneven land
(b) Clean, well-prepared moist seedbed
(c) Stony soil
(d) Deep mud

**Q107. The press wheel behind a furrow opener serves to:**

(a) Increase draft
(b) Firm soil around seed for better seed–soil contact and germination
(c) Meter the seed
(d) Mark the row

**Q108. Disc-type openers are preferred over shoe type when:**

(a) Field has heavy trash/residue
(b) Soil is clean and loose
(c) Speed is very low
(d) Seeds are very small

**Q109. Increasing the rake (approach) angle of a tine well beyond its optimum causes draft to:**

(a) Decrease
(b) Increase
(c) Remain constant
(d) Become zero

**Q110. Spring-loaded tines on cultivators are provided mainly to:**

(a) Increase working depth
(b) Protect against stones/obstructions by breakaway action
(c) Reduce cost
(d) Improve seed metering

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**Answers**

Q76 — (b) Horizontal component of pull in the direction of travel

Q77 — (b) 2819 N — *$D = 3000 \cdot \cos 20° = 3000 \times 0.9397 \approx 2819$ N; vertical component $3000 \cdot \sin 20° \approx 1026$ N (distractor a).*

Q78 — (c) Cross-sectional area of furrow

Q79 — (c) N/cm²

Q80 — (b) 98.1 kPa

Q81 — (c) 540 kgf — *$D$ = unit draft × furrow area × bottoms = 0.6 × (30×15) × 2 = 540 kgf.*

Q82 — (c) 10.8 hp — *$P = D \cdot v/75 = 540 \times 1.5/75 = 10.8$ hp ($\approx$ 7.95 kW).*

Q83 — (c) D·v/367

Q84 — (c) 13.5 kW — *$P(kW) = D(kN) \times v(m/s) = 9 \times 1.5 = 13.5$ kW.*

Q85 — (c) $\varepsilon abv^2$

Q86 — (b) Friction/rolling resistance due to implement weight

Q87 — (b) 4× — *Dynamic term $\propto v^2 \rightarrow 2^2 = 4\times$. This is why high-speed ploughing needs disproportionately more power.*

Q88 — (c) 1.0

Q89 — (a) ¼ width from landside, ½ depth

Q90 — (b) Centre of resistance and hitch point

Q91 — (b) The line of pull is laterally offset from the centre of resistance

Q92 — (b) Reduced penetration (plough runs shallow) — *Higher hitch → steeper pull line → larger upward component $P \cdot \sin\theta$ on the implement → reduced penetration.*

Q93 — (b) Added weight/ballast

Q94 — (c) Landside friction

Q95 — (b) Vertical force component

Q96 — (b) Dynamometer between drawbar and implement

Q97 — (b) Three-point hitch (strain-gauge) dynamometer

Q98 — (c) Higher
Q99 — (d) All of these
Q100 — (b) Working width
Q101 — (c) Transmission × tractive efficiency
Q102 — (c) 36 kW — *60 × 0.90 × 0.67 ≈ 36.2 kW — barely 60% of engine power reaches the drawbar on soil.*
Q103 — (b) Inverted-T
Q104 — (b) Precise, uniform seed placement and cut trash
Q105 — (b) Hard and dry soils
Q106 — (b) Clean, well-prepared moist seedbed
Q107 — (b) Firm soil around seed for better seed–soil contact and germination
Q108 — (a) Field has heavy trash/residue
Q109 — (b) Increase
Q110 — (b) Protect against stones/obstructions by breakaway action

## Sowing, fertilizer and inter-cultivation design — Topics 6–7 (Q111–Q145)

**Q111. The target cell fill for a precision planter plate is:**

(a) 50–60% (b) 70–80%
(c) 95–100% (d) Exactly 110%

**Q112. As plate (cell) linear speed increases, cell fill:**

(a) Increases (b) Decreases
(c) Constant (d) First falls then rises

**Q113. Barmington's data (Kepner Fig. 11.18) showed 100% cell fill for sugar-beet seed at a plate speed of about:**

(a) 10 fpm (b) 50 fpm
(c) 300 fpm (d) 500 fpm

**Q114. To plant at higher forward speeds with a plate meter, the designer should:**

(a) Speed up the plate (b) Use more cells per plate or a larger plate
(c) Reduce cells (d) Raise the hopper

**Q115. The metering-unit element that prevents multiple fill and seed breakage is the:**

(a) Knock-out (b) Cutoff device
(c) Agitator (d) Gate spring

**Q116. Positive removal of seed from cells is ensured by:**

(a) Gravity only (b) Knock-out/ejector devices
(c) Vacuum (d) Higher speed

**Q117. Conditions giving 95–100% cell fill also give:**

(a) Max damage (b) Most uniform spacing and minimum seed damage
(c) Higher draft (d) Poor spacing

**Q118. In fluted-roller metering, seed rate is chiefly controlled by:**

(a) Roller diameter (b) Exposed flute length
(c) Seed tube length (d) Hopper level

**Q119. Helical flutes (vs straight) on a fluted roller give:**

(a) Higher damage (b) Steadier, more uniform seed flow
(c) Lower capacity only (d) More wear

**Q120. Seed spacing delivered by a planter ($D_w$ wheel dia, $\sigma$ slip, $n_c$ cells, i plate/wheel ratio) is:**

(a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ (b) $n_c \cdot i/\pi D_w$

(c) $\pi D_w \cdot n_c \cdot \sigma$ (d) $D_w/n_c$

**Q121. Ground-wheel slip (skid) of a drill causes actual seed spacing to be:**

(a) Narrower than theoretical (b) Wider than theoretical

(c) Unchanged (d) Random

**Q122. Hopper walls must slope:**

(a) Less than angle of repose (b) Greater than the angle of repose of the material

(c) Exactly 30° (d) Any angle

**Q123. The angle between the two discs of a double-disc furrow opener is about:**

(a) 2–3° (b) 7–10°

(c) 20–25° (d) 40°

**Q124. Heavy down-force springs (50–150 kgf/opener) are characteristic of:**

(a) Broadcasting (b) No-till/zero-till openers

(c) Cup feed drills (d) Check-row planters

**Q125. A press wheel behind the opener serves to:**

(a) Mark rows (b) Firm soil around seed for better seed–soil contact

(c) Drive the meter (d) Cut trash

**Q126. Fertilizer metering is inherently difficult because fertilizers are:**

(a) Uniform and dry (b) Hygroscopic, corrosive, variable in flowability

(c) Dense only (d) Fibrous

**Q127. The feed used extensively on grain-drill fertilizing units is:**

(a) Vacuum disc (b) Star-wheel feed

(c) Cell plate (d) Spinning disc

**Q128. In star-wheel feed, application rate is adjusted by:**

(a) Fan speed (b) Gate opening and wheel size/speed

(c) Boom height (d) Nozzle size

**Q129. Auger (screw) feeder delivery rate is proportional to:**

(a) Screw speed and pitch volume (b) Hopper height

(c) Gate colour (d) Tube length

**Q130. The single-pass distribution pattern of a spinning-disc broadcaster is:**

(a) Rectangular (b) Triangular/oval, requiring 25–35% overlap

(c) Ring (d) Square

**Q131. In spinner spreading, fine particles:**

(a) Travel farthest (b) Fall short — causing pattern segregation

(c) Do not move (d) Bounce back

**Q132. Particle velocity leaving a spinner disc tip equals approximately:**

(a) $\omega \cdot r$ (b) $\omega/r$

(c) $r/\omega$ (d) $\omega^2 r$

**Q133. Recommended fertilizer band position relative to seed:**

(a) In contact (b) 5 cm to the side and 5 cm below

(c) 15 cm above (d) On the surface only

**Q134. Direct seed–fertilizer contact is avoided because:**

(a) Blocking tubes
(b) Salt injury to germinating seed
(c) Cost
(d) Corrosion of opener

**Q135. According to Kepner, the chief benefit of row-crop cultivation is:**

(a) Aeration
(b) Moisture mulch
(c) Weed control
(d) Root pruning

**Q136. The included angle of cultivator sweeps is generally:**

(a) 20–30°
(b) 60–90°
(c) 100–120°
(d) 150°

**Q137. Sweeps for dryland moisture conservation are designed with:**

(a) High lift and pitch
(b) Low pitch/lift — cut weeds without inversion
(c) Deep suction
(d) Narrow wings

**Q138. Near young row crops, cultivation depth should be:**

(a) 3–5 cm
(b) 10–12 cm
(c) 15–20 cm
(d) 25 cm

**Q139. Spring-trip cultivator standards are set to trip at about:**

(a) Working draft
(b) 1.5–2× working draft, with auto-reset
(c) 5× draft
(d) Any load

**Q140. Shields on row-crop cultivators:**

(a) Reduce draft
(b) Protect young plants from soil throw
(c) Meter seed
(d) Firm soil

**Q141. The parallel-linkage gang mounting on cultivators keeps:**

(a) Speed constant
(b) Sweep pitch/depth constant over uneven ground
(c) Row spacing fixed
(d) Draft zero

**Q142. The rotary hoe is used for:**

(a) Deep tillage
(b) Blind (pre/early post-emergence) shallow cultivation at high speed
(c) Puddling
(d) Ridging

**Q143. Number of drive-wheel revolutions to simulate area A in drill calibration:**

(a) $A/(\pi DWn)$
(b) $\pi DWn/A$
(c) $A \cdot \pi DWn$
(d) $\pi D/AWn$

**Q144. Acceptable coefficient of variation between seed-tube discharges of a drill is about:**

(a) ≤5–7%
(b) 15–20%
(c) 25–30%
(d) 50%

**Q145. A drill with 9 rows at 20 cm spacing, wheel dia 1.2 m: revolutions for 1/25 ha (400 m²) ≈**

(a) 25
(b) 59
(c) 100
(d) 236

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**Answers**

Q111 — (c) 95–100%
Q112 — (b) Decreases
Q113 — (b) 50 fpm
Q114 — (b) Use more cells per plate or a larger plate
Q115 — (b) Cutoff device

Q116 — (b) Knock-out/ejector devices
Q117 — (b) Most uniform spacing and minimum seed damage
Q118 — (b) Exposed flute length
Q119 — (b) Steadier, more uniform seed flow
Q120 — (a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ — *Option (a) mirrors the source notes; the physically correct form divides by the skid term: $s = \pi D_w/[n_c \cdot i \cdot (1 - \sigma)]$ (Topic 6a).*
Q121 — (b) Wider than theoretical — *A skidding ground wheel turns fewer times per metre, so seeds fall farther apart than the theoretical $\pi D/(n_c \cdot i)$.*
Q122 — (b) Greater than the angle of repose of the material
Q123 — (b) 7–10°
Q124 — (b) No-till/zero-till openers
Q125 — (b) Firm soil around seed for better seed–soil contact
Q126 — (b) Hygroscopic, corrosive, variable in flowability
Q127 — (b) Star-wheel feed
Q128 — (b) Gate opening and wheel size/speed
Q129 — (a) Screw speed and pitch volume
Q130 — (b) Triangular/oval, requiring 25–35% overlap
Q131 — (b) Fall short — causing pattern segregation
Q132 — (a) $\omega \cdot r$
Q133 — (b) 5 cm to the side and 5 cm below
Q134 — (b) Salt injury to germinating seed
Q135 — (c) Weed control
Q136 — (b) 60–90°
Q137 — (b) Low pitch/lift — cut weeds without inversion
Q138 — (a) 3–5 cm
Q139 — (b) 1.5–2× working draft, with auto-reset
Q140 — (b) Protect young plants from soil throw
Q141 — (b) Sweep pitch/depth constant over uneven ground
Q142 — (b) Blind (pre/early post-emergence) shallow cultivation at high speed
Q143 — (a) $A/(\pi DWn)$
Q144 — (a) ≤5–7%
Q145 — (b) 59 — *$N = A/(\pi \cdot D \cdot W \cdot n) = 400/(\pi \times 1.2 \times 0.20 \times 9) = 400/6.79 \approx 59$ revolutions.*

## Harvester and thresher design — Topic 8 (Q146–Q180)

**Q146. In a conventional mower cutter bar, knife-section pitch, guard spacing and stroke are all equal to:**

(a) 50.8 mm
(b) 76.2 mm
(c) 101.6 mm
(d) 152.4 mm

**Q147. Typical mower crank speed is 800–1000 rpm; the corresponding cutting strokes per minute are:**

(a) 400–500
(b) 800–1000
(c) 1600–2000
(d) 3200–4000

**Q148. Knife velocity in a crank-driven cutter bar is maximum at:**

(a) Start of stroke
(b) End of stroke
(c) Mid-stroke
(d) Uniform throughout

**Q149. For 76.2-mm stroke at 900 rpm crank speed, peak knife velocity is about:**

(a) 1.2 m/s
(b) 2.3 m/s
(c) 3.6 m/s
(d) 7.2 m/s

**Q150. A knife is in proper REGISTER when knife sections centre on the guards:**

(a) At mid-stroke
(b) At each end of the stroke
(c) Only while cutting
(d) At maximum velocity

**Q151. 'Lead' given to the outer end of a cutter bar compensates for:**

(a) Wheel slip
(b) Rearward flexing of the bar under cutting load
(c) Crop moisture
(d) Pitman wear

**Q152. Serrations on ledger plates mainly serve to:**

(a) Reduce weight
(b) Increase friction so stalks do not slip off the edges
(c) Sharpen the knife
(d) Reduce noise

**Q153. Rotary (disc) mowers cut by:**

(a) Shear between two edges
(b) Impact/free cutting using stalk inertia
(c) Tearing
(d) Sawing

**Q154. Impact cutting is possible only above a:**

(a) Minimum stalk diameter
(b) Minimum knife velocity
(c) Maximum moisture
(d) Minimum stubble height

**Q155. The wire-loop cylinder is characteristic of:**

(a) Maize sheller
(b) 'Hold-on' paddy threshers
(c) Groundnut decorticator
(d) Sunflower thresher

**Q156. A rasp-bar cylinder threshes mainly by:**

(a) Pure shear
(b) Impact plus rubbing of crop between bars and concave
(c) Centrifuging
(d) Suction

**Q157. Cylinder peripheral speed is given by:**

(a) $\pi DN/60$
(b) $60/\pi DN$
(c) $\pi D/60N$
(d) $DN/60$

**Q158. For wheat threshing at 28 m/s with a 55-cm cylinder, required rpm ≈**

(a) 490
(b) 720
(c) 972
(d) 1250

**Q159. Raising cylinder speed at fixed clearance generally:**

(a) Reduces threshing and damage
(b) Increases threshing completeness AND seed damage
(c) Increases losses only
(d) Has no effect

**Q160. Cylinder–concave clearance is normally:**

(a) Uniform
(b) Larger at feed end, tapering smaller at rear
(c) Smaller at feed end
(d) Zero at entry

**Q161. Visible seed damage rises approximately with:**

(a) $\sqrt{v}$
(b) $v$
(c) $v^2$
(d) $v^3$ of cylinder peripheral speed

**Q162. For threshing easily damaged podded seeds (e.g., beans), the gentlest unit is:**

(a) Spike-tooth cylinder
(b) Rubber rolls at 1.3–1.5 m/s
(c) Hammer beater
(d) High-speed rasp bar

**Q163. Threshing efficiency with 2% unthreshed grain is:**

(a) 2%
(b) 96%
(c) 98%
(d) 100%

**Q164. The axial-flow thresher achieves high efficiency with low damage because:**

(a) Very high speed
(b) Crop receives many gentler impacts while spiralling around the cylinder
(c) No concave
(d) Larger seeds

**Q165. Indian standard governing thresher feeding-chute safety is:**

(a) IS 4468
(b) IS 9129
(c) IS 6284
(d) IS 2062

**Q166. In Indian wheat threshers, fine bhusa is produced chiefly by:**

(a) Rubber rolls
(b) Beater/'drummy' and spike-tooth types
(c) Wire loop
(d) Stripper header

**Q167. In a combine, most of the threshed grain first separates at the:**

(a) Straw walkers
(b) Concave and grate
(c) Chaffer sieve
(d) Grain tank

**Q168. Straw walkers recover:**

(a) Unthreshed heads only
(b) Free grain remaining in the straw mat
(c) Chaff
(d) Weed seed

**Q169. The tailings elevator delivers tailings to the:**

(a) Grain tank
(b) Cylinder for re-threshing
(c) Straw spreader
(d) Fan

**Q170. The upper adjustable-lip sieve of the cleaning shoe is the:**

(a) Concave
(b) Chaffer
(c) Scalper
(d) Grader

**Q171. Excessive fan blast in the cleaning shoe causes:**

(a) Chaffy grain
(b) Free seed blown over the shoe (shoe loss)
(c) Unthreshed heads
(d) Header loss

**Q172. Reel index is the ratio of:**

(a) Reel diameter to cutting width
(b) Reel peripheral speed to forward speed
(c) Reel rpm to cylinder rpm
(d) Crop height to reel height

**Q173. Recommended reel index for combining standing cereals is about:**

(a) 0.5–0.8
(b) 1.0
(c) 1.25–1.5
(d) 2.5–3.0

**Q174. Too high a reel speed causes mainly:**

(a) Unthreshed loss
(b) Grain shattering at the header
(c) Walker loss
(d) Chaffy sample

**Q175. A pickup reel (spring-tine, cam-controlled) is used for:**

(a) Tall standing crop
(b) Lodged/down crop
(c) Root crops
(d) Green fodder

**Q176. At 5 km/h with reel index 1.3 and 1.0-m reel diameter, reel speed ≈**

(a) 15 rpm (b) 35 rpm
(c) 70 rpm (d) 140 rpm

**Q177. 'MOG' in combine capacity ratings means:**

(a) Moisture of grain (b) Material other than grain
(c) Mass of grain (d) Margin of grain

**Q178. Unthreshed seed discharged with straw from a combine is classified as:**

(a) Header loss (b) Cylinder loss
(c) Shoe loss (d) Rack loss

**Q179. In shattering-prone crops, the largest share of total combine loss is usually the:**

(a) Header (gathering) loss (b) Cylinder loss
(c) Shoe loss (d) Leakage

**Q180. Nearly all seed DAMAGE in a combine occurs at the:**

(a) Reel (b) Threshing (cylinder–concave) unit
(c) Grain tank (d) Straw spreader

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**Answers**

Q146 — (b) 76.2 mm

Q147 — (c) 1600–2000 — *Each crank revolution gives two cutting strokes (left + right), so 800–1000 rpm → 1600–2000 strokes/min.*

Q148 — (c) Mid-stroke

Q149 — (c) 3.6 m/s — *$\omega = 2\pi \times 900/60 = 94.2$ rad/s; $r = 38.1$ mm → $v_{max} = \omega r \approx 3.6$ m/s. Velocity is zero at stroke ends — hence cutting is timed near mid-stroke.*

Q150 — (b) At each end of the stroke

Q151 — (b) Rearward flexing of the bar under cutting load

Q152 — (b) Increase friction so stalks do not slip off the edges

Q153 — (b) Impact/free cutting using stalk inertia

Q154 — (b) Minimum knife velocity

Q155 — (b) 'Hold-on' paddy threshers

Q156 — (b) Impact plus rubbing of crop between bars and concave

Q157 — (a) $\pi DN/60$

Q158 — (c) 972 — *$N = 60v/\pi D = 60 \times 28/(3.1416 \times 0.55) = 972$ rpm.*

Q159 — (b) Increases threshing completeness AND seed damage

Q160 — (b) Larger at feed end, tapering smaller at rear

Q161 — (c) $v^2$ — *Kinetic energy of impact $\propto v^2$ — Kepner Fig. 17.7 shows damage climbing steeply while unthreshed loss falls as speed rises; the design compromise defines the working speed band.*

Q162 — (b) Rubber rolls at 1.3–1.5 m/s

Q163 — (c) 98% — *Threshing efficiency = 100 - % unthreshed = 98%.*

Q164 — (b) Crop receives many gentler impacts while spiralling around the cylinder

Q165 — (b) IS 9129

Q166 — (b) Beater/'drummy' and spike-tooth types

Q167 — (b) Concave and grate

Q168 — (b) Free grain remaining in the straw mat

Q169 — (b) Cylinder for re-threshing

Q170 — (b) Chaffer

Q171 — (b) Free seed blown over the shoe (shoe loss)

Q172 — (b) Reel peripheral speed to forward speed

Q173 — (c) 1.25–1.5

Q174 — (b) Grain shattering at the header

Q175 — (b) Lodged/down crop

Q176 — (b) 35 rpm — *Reel peripheral speed = 1.3 × (5/3.6) = 1.81 m/s; N = 60v/πD = 60×1.81/3.1416 ≈ 34.5 rpm.*

Q177 — (b) Material other than grain

Q178 — (b) Cylinder loss

Q179 — (a) Header (gathering) loss — *Header/gathering (shatter) loss commonly dominates in crops like sesame, mustard, soybean; it is controlled by reel setting, forward speed and timely harvest.*

Q180 — (b) Threshing (cylinder–concave) unit

## Controls, mechatronics, ML, IoT, robotics and drones — Topics 9, 10, 12 (Q181–Q215)

**Q181. Hydraulic systems work on:**

(a) Bernoulli's theorem
(b) Pascal's law
(c) Archimedes' principle
(d) Boyle's law

**Q182. Oil at 200 bar acts on a 50-cm² piston. The force developed is:**

(a) 10 kN
(b) 50 kN
(c) 100 kN
(d) 200 kN

**Q183. Hydraulic power in kW with p in bar and Q in L/min is:**

(a) pQ/60
(b) pQ/367
(c) pQ/600
(d) pQ/1000

**Q184. The hydraulic pump almost universally used on agricultural tractors is:**

(a) Centrifugal
(b) External gear
(c) Diaphragm
(d) Screw

**Q185. In an OPEN-centre hydraulic system, in neutral the pump flow:**

(a) Is blocked
(b) Returns freely to tank at low pressure
(c) Goes to the cylinder
(d) Stops

**Q186. A closed-centre load-sensing system uses a:**

(a) Fixed-displacement pump running at relief pressure
(b) Variable-displacement pump supplying flow on demand
(c) Hand pump
(d) Air compressor

**Q187. The relief valve in a hydraulic circuit:**

(a) Directs flow
(b) Limits maximum system pressure
(c) Filters oil
(d) Cools oil

**Q188. A single-acting hydraulic cylinder is lowered by:**

(a) Oil pressure
(b) Gravity/implement weight
(c) Spring only
(d) Air

**Q189. In a vacuum precision planter, seeds are held on the metering plate by:**

(a) Positive air pressure
(b) Negative pressure (suction)
(c) Glue
(d) Magnetism

**Q190. An air seeder distributes seed from a central hopper to openers by:**

(a) Gravity tubes
(b) Augers
(c) An air stream through manifolds
(d) Chain conveyors

**Q191. Compared with hydraulics, pneumatic systems have:**

(a) Higher force density
(b) Lower force density but cleaner operation
(c) Same force
(d) No actuators

**Q192. Mechatronics is the integration of:**

(a) Only mechanics and hydraulics
(b) Mechanics, electronics and computing/control
(c) Chemistry and biology
(d) Civil and electrical works

**Q193. The international standard for the tractor–implement electronic interface (ISOBUS) is:**

(a) ISO 730
(b) ISO 11783
(c) ISO 9001
(d) ISO 500

**Q194. ISOBUS communication is based on:**

(a) Bluetooth
(b) CAN bus
(c) Wi-Fi
(d) RS-232 only

**Q195. Draft on modern electro-hydraulic hitch tractors is sensed by:**

(a) Spring balance
(b) Load-sensing pins in the lower links
(c) Operator feel
(d) Fuel gauge

**Q196. Supervised machine learning requires:**

(a) Unlabelled data
(b) Labelled training data
(c) No data
(d) Only images

**Q197. The neural-network type dominating image-based weed detection is:**

(a) RNN
(b) CNN (convolutional)
(c) Perceptron only
(d) SOM

**Q198. 'See & Spray' technology saves chemicals by:**

(a) Diluting spray
(b) Spraying only where ML vision detects weeds
(c) Spraying at night
(d) Using more nozzles

**Q199. The IoT protocol designed for kilometres of range at very low power is:**

(a) Bluetooth
(b) LoRaWAN
(c) USB
(d) NFC

**Q200. NDVI is computed as:**

(a) (R-NIR)/(R+NIR)
(b) (NIR-R)/(NIR+R)
(c) NIR×R
(d) NIR/G

**Q201. RTK-GNSS auto-steer gives pass-to-pass accuracy of about:**

(a) 2–3 m
(b) 30–50 cm
(c) 2–3 cm
(d) 2–3 mm

**Q202. The robot component that actually grips/cuts the fruit is the:**

(a) Manipulator base
(b) End-effector
(c) Encoder
(d) Gantry

**Q203. Edge computing in IoT means processing data:**

(a) Only in cloud
(b) On or near the sensing device
(c) On paper
(d) After harvest

**Q204. Under Drone Rules 2021, agricultural spray drones (typically 2–25 kg) fall in the class:**

(a) Nano
(b) Micro
(c) Small
(d) Medium

**Q205. A NANO drone weighs:**

(a) <250 g (b) <2 kg

(c) <25 kg (d) <150 kg

**Q206. India's online platform for drone registration, UIN and airspace maps is:**

(a) UMANG (b) Digital Sky (DGCA)

(c) DigiLocker (d) AGMARKNET

**Q207. In a GREEN zone, drones may operate without permission up to:**

(a) 60 m (b) 120 m (400 ft)

(c) 500 m (d) Unlimited

**Q208. A Remote Pilot Certificate is NOT required for:**

(a) Small drones (b) Medium drones

(c) Nano and non-commercial micro drones (d) Any drone

**Q209. Typical drone ULV spray application rate is:**

(a) 2–4 L/ha (b) 20–40 L/ha

(c) 200–400 L/ha (d) 500–600 L/ha

**Q210. Recommended spraying height of an agri drone above the crop canopy is about:**

(a) 0.2–0.5 m (b) 2–3 m

(c) 8–10 m (d) 15–20 m

**Q211. Drone spraying should be avoided when wind speed exceeds about:**

(a) 2 km/h (b) 10 km/h

(c) 30 km/h (d) 50 km/h

**Q212. Agricultural drones are commonly powered by:**

(a) Lead-acid batteries (b) LiPo (lithium-polymer) batteries

(c) NiCd (d) Dry cells

**Q213. Under SMAM, ICAR institutes/KVKs/SAUs get drone assistance of:**

(a) 40% (b) 50%

(c) 75% (d) 100% up to ₹10 lakh

**Q214. Namo Drone Didi provides drones to women SHGs at a subsidy of:**

(a) 40% up to ₹4 lakh (b) 50% up to ₹5 lakh

(c) 80% up to ₹8 lakh (d) 100%

**Q215. For large-area aerial survey/imaging with long endurance, the preferred platform is:**

(a) Quadcopter (b) Fixed-wing drone

(c) Tethered balloon (d) Hexacopter

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**Answers**

Q181 — (b) Pascal's law

Q182 — (c) 100 kN — *$F = pA = 200\times10^5$ Pa × $50\times10^{-4}$ m² = 100,000 N = 100 kN — hydraulic force multiplication.*

Q183 — (c) pQ/600 — *P(kW) = p(bar)×Q(L/min)/600; e.g., 180 bar × 40 L/min → 12 kW.*

Q184 — (b) External gear

Q185 — (b) Returns freely to tank at low pressure

Q186 — (b) Variable-displacement pump supplying flow on demand

Q187 — (b) Limits maximum system pressure

Q188 — (b) Gravity/implement weight

Q189 — (b) Negative pressure (suction) — *Vacuum plate = singulation by suction (precision planting); air seeder = pneumatic bulk distribution — two different uses of air.*

Q190 — (c) An air stream through manifolds — *Vacuum plate = singulation by suction; air seeder = pneumatic bulk distribution — two different uses of air.*

Q191 — (b) Lower force density but cleaner operation

Q192 — (b) Mechanics, electronics and computing/control

Q193 — (b) ISO 11783

Q194 — (b) CAN bus

Q195 — (b) Load-sensing pins in the lower links

Q196 — (b) Labelled training data

Q197 — (b) CNN (convolutional)

Q198 — (b) Spraying only where ML vision detects weeds

Q199 — (b) LoRaWAN

Q200 — (b) (NIR−R)/(NIR+R) — *Healthy vegetation reflects NIR strongly and absorbs red → NDVI near +0.6–0.9; stressed crop lower (basis of drone crop monitoring, Unit 10).*

Q201 — (c) 2–3 cm

Q202 — (b) End-effector

Q203 — (b) On or near the sensing device

Q204 — (c) Small — *Drone Rules 2021 — nano <250 g, micro ≤ 2 kg, small ≤ 25 kg, medium ≤ 150 kg, large >150 kg.*

Q205 — (a) <250 g

Q206 — (b) Digital Sky (DGCA)

Q207 — (b) 120 m (400 ft) — *Green zone ceiling 120 m.*

Q208 — (c) Nano and non-commercial micro drones — *RPC exemption only for nano and non-commercial micro drones.*

Q209 — (b) 20–40 L/ha — *Closest option — the 2023 crop-specific SOP sets about 20–25 L/ha by crop stage (Topic 12a).*

Q210 — (b) 2–3 m — *Closest option — summaries of the 2023 SOP give about 1.5–2.5 m above the canopy (3–4 m in sugarcane).*

Q211 — (b) 10 km/h

Q212 — (b) LiPo (lithium-polymer) batteries

Q213 — (d) 100% up to ₹10 lakh — *SMAM slabs — 100%/₹10 lakh (ICAR/KVK/SAU), 75% FPO demos, CHC 40%/₹4 lakh (50%/₹5 lakh agri graduates), individuals 40–50%.*

Q214 — (c) 80% up to ₹8 lakh — *Namo Drone Didi — 80%/₹8 lakh, 15,000 women SHGs, ₹1,261 crore (2023-24 to 2025-26).*

Q215 — (b) Fixed-wing drone